

Chapter 4

Evidence-Aware Evaluation of Question Decomposition

Introduction

Chapter 3 examined how compositional difficulty is represented before a model answers a question. This chapter studies the retrieval process that follows a decision to decompose. Its objective is to measure whether the resulting subquestions retrieve the evidence needed for the original question, and how deeply each relevant retrieval list must be searched.

A decomposition changes several parts of the computation at once. It creates additional retrieval queries, assigns information needs to those queries and passes intermediate answers between them. Final-answer accuracy combines these effects. Passage recall measures evidence availability but leaves the assignment of that evidence to individual subquestions unspecified. Evaluating the decomposition therefore requires three separate quantities: complete evidence availability, evidence ownership and successful answer execution.

The chapter develops an evaluation based on a matrix of evidence ranks and a learned ownership mask. The mask identifies which supporting passages each subquestion is intended to retrieve. A maximum over the selected cells measures the depth required to satisfy those assignments. A separate union measurement records whether the evidence was retrieved anywhere in the execution. The ownership model is evaluated on reference-style decompositions whose assignments are known. Retrieval and answer results are then compared on a 100-question MuSiQue panel. Finally, fixed-list budget replay distinguishes per-query depth from total passage expenditure. These measurements provide the objective and diagnostic information used for prompt optimisation in Chapter 5.

4.1 Problem Formulation

4.1.1 A Decomposition as an Executable Sequence

Let q be a question with reference answer y and annotated supporting passages $E(q) = \{e_1, \dots, e_h\}$. MuSiQue supplies both supporting passages and a reference decomposition into connected single-hop questions [1]. A generated decomposition is an ordered sequence $U(q) = (u_1, \dots, u_A)$. An item u_a may contain a reference to the answer of an earlier item. The execution substitutes the predicted earlier answer, retrieves passages for the resulting

query $\tilde{u}_a$ and produces an intermediate answer $\hat{y}_a$:

$$\tilde{u}_a = \text{substitute}(u_a, \hat{y}_1, \dots, \hat{y}_{a-1}), \quad \hat{y}_a = \text{reader}(\tilde{u}_a, R_a[1 : k]). \quad (4.1)$$

Here $R_a = (d_{a1}, d_{a2}, \dots)$ is the ranked passage list returned for $\tilde{u}_a$, and k is the number of passages supplied to the reader. Backward references make the execution sequential even when the decomposition contains conceptually independent branches. The retained implementation returns $\hat{y}_A$ as the final answer. Each step receives its own retrieved context; the final step has no additional synthesis call over the union of all earlier passages. One such step, a resolved query with its retrieval and its reader call, is an arm of the execution.

The number of generated subquestions A may differ from the reference hop count h . One subquestion can combine several reference information needs. Several subquestions can address the same need. The evaluation therefore permits many-to-many assignments instead of requiring an alignment by position.

4.1.2 Availability, Assignment and Execution

The three evaluation objects are defined as follows.

Availability. Every annotated supporting passage occurs in at least one retrieved prefix.

Assignment. Every supporting passage has an identified owner, and each owning subquestion retrieves the passages assigned to it.

Execution. The reader produces the required intermediate answers and the final answer matches the reference under the specified answer metric.

These properties answer different questions. Evidence retrieved for a later subquestion may have been unavailable when an earlier answer was produced. Conversely, a model may answer correctly from a passage outside the annotated evidence set or from parametric knowledge. The measurements below concern recovery of benchmark evidence and reference-answer accuracy. Their relation is examined empirically rather than imposed by definition.

4.2 Retrieval Depth and Evidence Responsibility

4.2.1 The Evidence-Rank Matrix

For a recorded execution, define an $A \times h$ matrix

$$\rho_{ai} = \min\{r : d_{ar} = e_i\}. \quad (4.2)$$

The evaluation retains the first $K = 200$ ranked passages per subquestion. When e_i is absent from that prefix, its recorded rank is $+\infty$. This value denotes censoring at K , including evidence that might occur deeper in the full retrieval list. Passage matching uses the benchmark passage identity, determined from its title and text.

The matrix records where each required passage occurs for each query. It does not determine which query was responsible for that passage. That second object is introduced below.

4.2.2 Union Depth

The union depth is the smallest common prefix length that makes all annotated evidence available somewhere in the retrieval lists:

$$D_{\text{union}}(q) = \max_{1 \leq i \leq h} \min_{1 \leq a \leq A} \rho_{ai}. \quad (4.3)$$

Consequently, $D_{\text{union}}(q) \leq k$ exactly when $E(q) \subseteq \bigcup_a R_a[1 : k]$. The inner minimum allows one successful retriever to satisfy the availability requirement for a passage. The outer maximum requires every annotated passage to be present.

This definition is appropriate for measuring a pooled evidence collection. In the sequential execution of Equation 4.1, however, a passage can appear in the pooled collection without reaching the particular reader call that needed it. Union depth is therefore reported alongside an assignment-aware measurement.

4.2.3 Ownership-Masked Depth

Let $M \in \{0, 1\}^{A \times h}$ be an ownership mask, with $M_{ai} = 1$ when subquestion a is assigned responsibility for passage e_i . Define

$$D_M(q) = \begin{cases} \max_{(a,i): M_{ai}=1} \rho_{ai}, & \text{if every column of } M \text{ contains an owner,} \\ +\infty, & \text{otherwise.} \end{cases} \quad (4.4)$$

An assigned cell whose evidence is absent from the retained list also makes $D_M(q) = +\infty$. The evaluation distinguishes this retrieval censoring from an unowned column, which indicates an uncovered assignment under the predicted mask.

The row and column structure has a direct interpretation. A zero row is an idle subquestion with respect to the annotated evidence. A row with several ones assigns several evidence items to one subquestion. A zero column leaves an evidence item unowned. A column with several ones assigns that evidence to several subquestions. Idle rows contribute no retrieval obligation to Equation 4.4; their generation and reading costs remain part of the executed system.

Multiple ownership is interpreted conjunctively: each selected owner must retrieve the passage. A rule that accepts retrieval by any selected owner would use a minimum over owners and would measure a different execution requirement. The present rule preserves the interpretation of each positive cell as a specific obligation of a specific reader call.

For comparison, the all-cells maximum is

$$D_{\text{all}}(q) = \max_{a,i} \rho_{ai}. \quad (4.5)$$

It requires every subquestion to retrieve every supporting passage. Whenever the mask assigns every evidence item, the three quantities satisfy

$$D_{\text{union}}(q) \leq D_M(q) \leq D_{\text{all}}(q). \quad (4.6)$$

The first inequality follows because each selected rank is at least the smallest rank in its column. The second follows because the masked maximum ranges over a subset of all cells. A zero column is handled separately by the censoring rule in Equation 4.4.

For example, consider

$$\rho = \begin{pmatrix} 1 & 80 \\ 40 & 2 \end{pmatrix}, \quad M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \quad (4.7)$$

Both union depth and ownership-masked depth equal 2, while the all-cells maximum equals 80. Each subquestion retrieves its own assigned passage near the beginning of its list. Requiring the off-diagonal passages changes the question being evaluated. With a single full-question query, the control assigns all evidence to that query, so the three depth definitions coincide.

4.2.4 Coverage and an Aggregate Depth Score

For a panel of n questions, complete-evidence coverage at depth k is

$$C_D(k) = \frac{1}{n} \sum_{j=1}^n \mathbf{1}\{D(q_j) \leq k\}, \quad (4.8)$$

where D is either D_{union} or D_M . Coverage is a question-level complete-set measurement. A question missing one required passage receives the same coverage value as a question missing several, while the retained rank matrix identifies the individual missing items.

Prompt optimisation also requires a scalar objective. The retained objective assigns logarithmic credit to finite depths:

$$v_K(d) = \begin{cases} 1 - \frac{\log d}{\log K}, & 1 \leq d \leq K, \\ 0, & d = +\infty, \end{cases} \quad J_D = \frac{100}{n} \sum_{j=1}^n v_K(D(q_j)). \quad (4.9)$$

The score is 100 when all required evidence appears at rank 1 and 0 when every question is censored or reaches depth K . Questions with unowned evidence remain in the denominator. Reporting $C_D(K)$ beside J_D distinguishes incomplete retrieval from complete retrieval at the search limit. The score summarises complete coverage and depth. Arm count and model tokens are recorded separately because Equation 4.9 does not price them.

4.3 Learning the Ownership Mask

4.3.1 Supervision from Reference Decompositions

MuSiQue associates each reference subquestion with a supporting passage. Substituting the reference intermediate answers resolves its dependencies and provides labelled subquestion–passage pairs. The associated passage is a positive ownership example; the other supporting passages of the same original question provide negative examples. These labels encode the benchmark’s reference assignment. An alternative valid decomposition can have a different assignment.

Training uses a curated set of 6 847 MuSiQue training questions. The resulting training matrix contains 36 694 cells, of which 15 452 are positive. The held-out control contains 100 questions with reference-style subquestions and known reference assignments. The retained masks provide the predictions evaluated below; the archived template history

is only partially documented. The control contains 48 two-hop, 30 three-hop and 22 four-hop questions.

The held-out question identifiers and exact resolved-subquestion–passage pairs are absent from the training set. Within the training set, questions sharing an exact repeated pair are assigned together to prevent that pair from appearing in both fitting and evaluation folds.

4.3.2 Textual and Contextual Features

For each candidate cell, a 193-dimensional feature vector describes the relation between the resolved subquestion and the supporting passage. The features combine passage-level and sentence-level ColBERT scores, lexical similarity, token alignment, title overlap and features of the subquestion before intermediate-answer substitution. ColBERT supplies contextual token representations with a late interaction score [2].

Absolute support and relative support are both represented. Background normalisation compares semantic scores against a fixed pool of 300 passages drawn from training data. Row features compare one subquestion’s support for different evidence passages. Column features compare different subquestions’ support for the same passage. Agreement features record whether different similarity measures identify the same candidate.

These comparisons are relevant to ownership: two passages may share an entity, while only one answers the relation requested by the subquestion. They also permit a cell to receive low support even when it is the best option in its row. The predictor can consequently leave a row or column empty. Retrieval rank, measured depth, reference hop position and the diagonal reference label are excluded from the feature vector. The predictor is fitted to textual assignment labels rather than to the retrieval outcomes that it will help evaluate.

The evaluation still supplies the annotated evidence passages as candidates. Thus the mask is an offline attribution instrument with known evidence to assign. Using text-based features does not by itself provide the complete evidence set for a new question at deployment.

4.3.3 Fitting, Threshold Selection and Ensemble Decision

Five grouped folds are constructed while balancing reference hop cohorts. Each rotation uses three folds to fit a weighted logistic model, one disjoint fold to select its threshold and one fold for evaluation. Regularisation is selected within the fitting portion. Standardisation parameters are estimated from the same fitting data. The retained configuration gives positive ownership examples weight 3.

For model b , let $p_b(a, i)$ be its predicted ownership score. Its threshold is obtained from the lower tail of scores assigned to positive calibration cells. If those scores in ascending order are $p_{(1)}, \dots, p_{(m)}$, the rule is

$$t_b = p_{(\lfloor (m+1)\alpha \rfloor)}, \quad \alpha = 0.01, \quad (4.10)$$

with threshold $-\infty$ when the indicated index is zero. The final mask uses a majority vote:

$$\widehat{M}_{ai} = \mathbf{1} \left\{ \sum_{b=1}^5 \mathbf{1}\{p_b(a, i) \geq t_b\} \geq 3 \right\}. \quad (4.11)$$

The lower-quantile threshold favours retaining true ownership assignments. Here α specifies the threshold-selection rule. Empirical recall, precision and complete-mask agreement quantify the resulting five-model procedure. Recall and precision are computed over mask cells pooled across questions, complete-mask agreement is the fraction of questions whose predicted mask matches the reference in every cell, and an arm row is exact when the prediction matches the reference in every cell of that row. A per-cell quantile rule and a majority vote require additional assumptions and analysis to yield a simultaneous guarantee for every assignment in a question. The results in this chapter are held-out measurements of the specified ensemble.

4.3.4 How Assignment Error Changes the Depth Measurement

Ownership prediction affects the measurement itself. Adding a positive cell adds a retrieval obligation and can increase the masked maximum. Removing a positive cell can reduce that maximum if the corresponding passage retains another owner. Removing its last owner instead censors the question. Mask errors can therefore move measured depth in either direction.

This dependence motivates evaluation at three levels: individual cells, complete arm rows and complete question masks. It also motivates freezing the attribution model when comparing decomposition prompts. A change in attribution must be distinguished from a change in the underlying retrieved lists.

4.4 Experimental Protocol

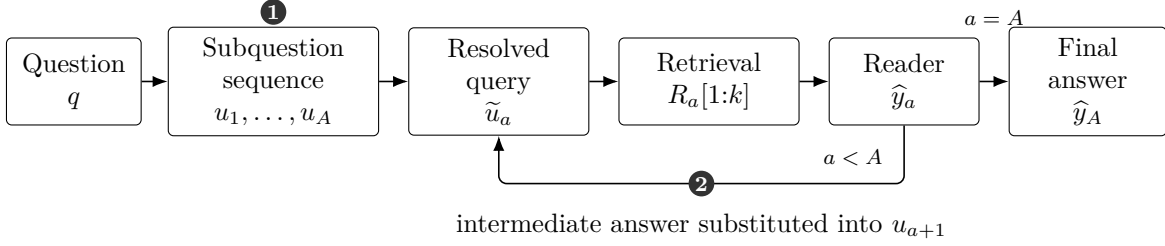
4.4.1 Systems and Retained Executions

The retrieval study uses the same 100 MuSiQue questions across the compared configurations. Its two language-model settings use GPT-4o-mini and Llama-3.3-70B-Instruct respectively for decomposition and reading. Retrieval uses HippoRAG 2 with NV-Embed-v2 embeddings and the model-specific graph and reranking configuration. HippoRAG 2 combines passage information with graph retrieval based on personalised PageRank [3]. Each reader call receives five passages; the retained rankings extend to 200 passages for depth analysis.

Three generated-decomposition prompts are compared with direct full-question retrieval. The baseline decomposition prompt serves as the reference generated policy, while the atomic-query prompt requests atomic single-fact queries. Both prompts use the same three demonstrations. The extended-instruction prompt uses a more detailed instruction specification without those demonstrations. These are different prompting configurations, so their comparison measures the full prompting choice rather than instruction length alone.

Two GPT-4o-mini controls use the reference decomposition. The first substitutes gold intermediate answers into downstream queries. The second substitutes the reader’s predicted intermediate answers. Both retain reference subquestions; the gold-answer condition isolates the effect of supplying correct intermediate bindings within this pipeline.

Figure 4.1 shows the execution shared by these configurations and states what each condition supplies at the two points that the comparison varies.



Condition	① Subquestion sequence	② Intermediate bindings
Direct	None: the full question is the only query	None: one retrieval and one reader call
Generated plan	Prompt-generated (baseline, atomic-query, extended-instruction)	Reader's predicted intermediate answers
Reference plan (predicted)	MuSiQue reference decomposition	Reader's predicted intermediate answers
Reference plan (gold)	MuSiQue reference decomposition	Gold intermediate answers (oracle-assisted)

Figure 4.1: Execution conditions in the retrieval study. The diagram is the execution shared by every condition, with the two marked points that the comparison varies: the source of the subquestion sequence and the source of the intermediate answers substituted into later subquestions. The table states what each condition supplies at those two points. Direct execution has neither, and the gold-binding condition is oracle-assisted.

4.4.2 Metrics and Comparison Units

Answer quality is measured by exact match (EM) and token F1 against the reference answer and its accepted aliases, under the benchmark’s answer normalisation: lowercasing, removal of articles and punctuation, and collapse of whitespace [1]. Let $t(\cdot)$ denote the token multiset of a normalised string, and let $\mathcal{A}(y)$ denote the reference answer y together with its accepted aliases. Then

$$\text{EM}(\hat{y}, y) = \mathbf{1}\{\hat{y} \in \mathcal{A}(y)\}, \quad \text{F1}(\hat{y}, y) = \max_{y' \in \mathcal{A}(y)} \frac{2|t(\hat{y}) \cap t(y')|}{|t(\hat{y})| + |t(y')|}, \quad (4.12)$$

where all compared strings are normalised and $\cap$ is the multiset intersection. Both quantities are averaged over the panel; EM is reported as a percentage and F1 on a 0–100 scale.

Retrieval is measured by complete-evidence coverage (Equation 4.8), union depth (Equation 4.3) and ownership-masked depth (Equation 4.4), summarised by the depth score J_D of Equation 4.9. The number of generated arms, retrieved passage slots and recorded total model tokens describe resource use. A passage slot is one passage delivered to one reader call. Per-arm passage counts therefore charge duplicate passages again when they are sent to separate reader calls, while unique passage counts describe the pooled collection. Recorded tokens are the prompt and completion tokens of the decomposition and reading calls of a question.

The reference-answer control records one token total across several execution branches. Per-branch costs are therefore unavailable. Generated and direct runs have separate recorded totals, which are reported per question.

Comparisons pair the same questions across configurations. Where intervals are reported, they are percentile intervals from 10 000 paired question-level bootstrap resamples.

They describe variation within this development panel. Several prompt and attribution configurations were examined on the panel, so these intervals are descriptive rather than a correction for adaptive model selection. The reference-style ownership control was kept separate from weight fitting and threshold calibration; its role is a structured assignment control rather than an evaluation of arbitrary generated plans.

4.5 Results

4.5.1 Ownership Recovery on Reference-Style Arms

Table 4.1 reports the retained ownership ensemble on the reference-style control. Of 274 reference owner cells, 269 are recovered. Seven non-owner cells are selected. Precision is 97.46% and recall is 98.18%. The ensemble recovers all assignments exactly on 92 of the 100 questions. It recovers 264 of the 274 complete arm rows, which is a different unit from complete question-mask agreement.

Table 4.1: Ownership prediction on the reference-style decomposition control. Cell metrics use 814 subquestion–passage pairs. Complete-mask agreement is measured over the 100 questions.

Measurement	Value
True positives / false negatives	269 / 5
False positives / true negatives	7 / 533
Precision	97.46%
Recall	98.18%
Cell accuracy	98.53%
AUROC of mean model scores	0.9977
Exact arm rows	264/274
Exact question masks	92/100

The cohort breakdown in Table 4.2 shows the distribution of the twelve cell errors. Four-hop questions contain three of the five missed owners. Because several cells belong to the same question, cell totals are reported with their question counts rather than interpreted as independent question outcomes.

Table 4.2: Ownership errors by reference hop count. TP, FN, FP and TN count individual cells, while n counts questions.

Reference hops	n	Cells	TP	FN	FP	TN
2	48	192	95	1	4	92
3	30	270	89	1	0	180
4	22	352	85	3	3	261

On the gold-answer retrieval traces, the exact reference mask yields complete assigned-evidence coverage of 100% at depth 200 and a score of 86.47. Using the learned ownership mask gives 97% coverage and a score of 84.84. This comparison isolates the effect of replacing the reference assignment with its prediction on those fixed lists.

4.5.2 Sensitivity to the Attribution Model

Table 4.3 scores the same retrieval traces under two attribution models: an earlier predictor built on a different feature set, and the retained model of Section 4.3. The baseline generated decomposition receives scores of 52.32 and 72.19 respectively. The analogous atomic-query values are 52.81 and 66.64. These changes arise from the assigned cells, with the passages and ranks held fixed. On the reference arms with gold bindings the two models differ by 24.98 points, while replacing the retained prediction with the exact reference assignment changes the score by 1.63. The masked-depth measurement is therefore more sensitive to the choice of attribution model than to the residual error of the retained one.

Table 4.3: Masked-depth measurements under two attribution models on fixed GPT-4o-mini retrieval traces. Coverage is complete assigned-evidence coverage at $K = 200$; score is J_{D_M} from Equation 4.9. The passages and their ranks are identical across rows, so only the assigned cells differ. The last row uses the exact reference assignment.

Decomposition	Attribution	Coverage (%)	Score
Baseline	Earlier model	78	52.32
Baseline	Retained model	87	72.19
Atomic-query	Earlier model	77	52.81
Atomic-query	Retained model	81	66.64
Reference, gold bindings	Earlier model	88	59.86
Reference, gold bindings	Retained model	97	84.84
Reference, gold bindings	Reference assignment	100	86.47

The reference-style control supports the use of the retained predictor for recovering known assignments. For generated arms, assignments can legitimately be fused or shared. The score differences in Table 4.3 therefore establish sensitivity to attribution rather than a labelled accuracy comparison on generated plans. Prompt comparisons in Chapter 5 keep the current ownership model fixed.

4.5.3 Evidence Availability and Final Answers

For readability, $C_U(k)$ denotes the union coverage $C_{D_{\text{union}}}(k)$ in the result tables. Table 4.4 reports the direct and generated-decomposition executions. With GPT-4o-mini, the baseline decomposition raises complete union coverage at five passages per arm from 45% to 76%. EM changes from 40% to 52% and F1 from 52.64 to 59.00. With Llama-3.3-70B, complete union coverage changes from 46% to 79%, while EM changes from 42% to 43% and F1 from 52.30 to 51.37.

The paired bootstrap interval for the baseline decomposition’s EM change is $[2, 22]$ percentage points for GPT-4o-mini and $[-8, 10]$ for Llama-3.3-70B. These measurements show why retrieval and answer metrics are both needed. Improved pooled availability can coexist with a small change in final answers, because the sequence must still use the retrieved evidence and propagate useful intermediate states.

The per-arm comparison also changes the total retrieval expenditure. The baseline GPT-4o-mini decomposition retrieves a mean of 13.95 passage slots across its reader calls, compared with 5 for direct retrieval. Its mean pooled collection contains 11.35 distinct passages. Thus the coverage increase in Table 4.4 combines a change in query

Table 4.4: Recorded execution results on 100 MuSiQue questions. EM and F1 are percentages. $C_U(5)$ is complete union coverage at five passages per arm. Arms and tokens are per-question means. Retrieved evidence and final answers are separate outcomes.

Model	Prompt	EM	F1	$C_U(5)$	Arms	Tokens
GPT-4o-mini	Direct	40	52.64	45	1.00	4 369
	Baseline	52	59.00	76	2.79	12 763
	Atomic-query	52	59.66	72	2.66	11 898
	Extended-instruction	50	60.39	74	2.61	13 881
Llama-3.3-70B	Direct	42	52.30	46	1.00	4 530
	Baseline	43	51.37	79	3.00	14 069
	Atomic-query	45	53.54	78	3.22	14 789
	Extended-instruction	47	55.94	75	2.84	15 271

formulation with more passage slots and more reader calls. Section 4.6 compares fixed lists at a common total passage budget.

4.5.4 The Role of Intermediate Bindings

The two reference-decomposition controls use the same subquestion templates and reader but change the answers substituted into later queries. Table 4.5 reports their outcomes. Gold substitution yields EM 64% versus 49% under predicted substitution, with a paired bootstrap interval of $[8, 22]$ percentage points for the difference. Complete union coverage at five passages per arm changes from 80% to 92%.

Table 4.5: Reference-decomposition controls with GPT-4o-mini. Gold bindings supply reference intermediate answers when constructing downstream queries. Predicted bindings use the reader’s earlier outputs.

Intermediate substitution	EM (%)	F1	$C_U(5)$ (%)
Predicted answers	49	57.50	80
Gold answers	64	70.73	92

The intervention changes downstream query text and consequently can change both retrieval and answer generation. Its effect includes error propagation through those two stages. Supplying gold intermediate answers defines an oracle-assisted control. The measured gap identifies a role for intermediate state quality within the reference-chain pipeline; it is not a gain achieved by an operational binding policy.

4.6 Depth Under a Total Passage Budget

4.6.1 Equal-Allocation Replay

A common per-arm depth gives more total passage slots to longer decompositions. To separate these quantities, consider a per-question ceiling $B \leq AK$ distributed among the A recorded retrieval lists. Equal allocation uses the whole allowance and assigns

$$k_a(B) = \left\lfloor \frac{B}{A} \right\rfloor + \mathbf{1}\{a \leq B \bmod A\}, \quad \sum_{a=1}^A k_a(B) = B. \quad (4.13)$$

The remainder is allocated in execution order, a tie-break over at most $A - 1$ slots that favours the earlier arms whose answers resolve later queries. A passage slot is charged each time it occurs in an arm prefix, including duplicates across arms. Complete evidence availability is then evaluated in $\bigcup_a R_a[1 : k_a(B)]$.

This allowance counts selected passage slots. Equal slot counts can produce different numbers of unique passages and different token totals, since passages can recur across arms and vary in length. Retrieving the ranked lists, producing the queries and reading the contexts incur separate computational costs.

These are fixed-list replays. The original intermediate answers and later queries were generated using five passages per reader call. Changing a prefix in the replay leaves those earlier answers and later queries unchanged. The result measures the coverage of the recorded lists at a specified slot budget. An end-to-end policy evaluated at that budget would regenerate intermediate answers and dependent queries under its actual contexts.

Per-question and average budgets. Under a hard ceiling B for each question, the largest feasible common depth within a question is $\lfloor B/A(q) \rfloor$. Complete-evidence availability is non-decreasing in every prefix length, so this choice weakly dominates smaller equal depths. Equation 4.13 also spends the integer remainder. A single depth shared across the entire panel is a weaker comparator when arm counts vary: feasibility limits it to $\lfloor B/\max_q A(q) \rfloor$. Unequal within-question allocation can improve on the balanced comparator by placing more slots on deeper arms. An average budget instead permits transfers between questions, including transfers that retain positive allocations for every question. That is a different resource constraint from the per-question comparisons here.

4.6.2 A Fixed-List Oracle Allocation

The rank matrix also permits an exact comparison with the least expensive prefix allocation for its recorded lists:

$$B^*(q) = \min_{k_1, \dots, k_A \in \{0, \dots, K\}} \left\{ \sum_a k_a : E(q) \subseteq \bigcup_a R_a[1 : k_a] \right\}. \quad (4.14)$$

The value is $+\infty$ when the retained lists jointly miss annotated evidence. For finite cases, it can be computed by dynamic programming over subsets of evidence. Let $S_a(k)$ denote the evidence indices found in arm a 's first k passages. Starting from $F_0(\emptyset) = 0$, the update is

$$F_a(T \cup S_a(k)) \leftarrow \min\{F_a(T \cup S_a(k)), F_{a-1}(T) + k\}. \quad (4.15)$$

Unreachable states have value $+\infty$, and $B^*(q) = F_A(\{1, \dots, h\})$. Only zero and prefix lengths at which an additional evidence item appears need to be considered: a longer prefix with the same evidence set adds cost without changing that state. With at most four annotated hops in this panel, the subset state space is small.

The oracle uses gold evidence identities to choose its allocation. Moreover, all recorded queries are available even when their selected prefix length is zero. The objective charges passage slots but leaves the generation of those queries and their intermediate states outside the optimisation. It is an exact minimum for the fixed-list problem and an optimistic reference for a system that must construct and execute the queries. It is not an achieved runtime or token saving.

4.6.3 Allocation Headroom on the Recorded Lists

The budget $B = 15$ is close to the mean of 13.95 passage slots recorded for the baseline GPT-4o-mini decomposition, so the replay compares configurations at approximately the cost of that decomposition. At $B = 15$, equal allocation recovers complete evidence for 76 questions under the baseline GPT-4o-mini decomposition and 78 under its Llama counterpart. The corresponding oracle allocations reach 85 and 87; an oracle entry counts the questions whose minimum feasible budget $B^*(q)$ in Equation 4.14 is at most 15. Table 4.6 reports the same comparison for the other prompts and for direct retrieval, which are the configurations available without a reference decomposition. Direct retrieval has a single arm, so its two allocations coincide by construction, and its entries give that one list 15 passages rather than the five used in execution.

Table 4.6: Complete evidence coverage (%) at a total budget of 15 passage slots on fixed recorded lists. The oracle chooses prefixes using gold evidence and is a reference allocation rather than a deployed policy. Each row contains 100 questions.

Model	Prompt	Equal allocation	Oracle allocation
GPT-4o-mini	Direct	66	66
	Baseline	76	85
	Atomic-query	73	85
	Extended-instruction	75	83
Llama-3.3-70B	Direct	65	65
	Baseline	78	87
	Atomic-query	75	88
	Extended-instruction	75	85

The baseline decompositions retain a coverage advantage over direct retrieval in this replay after the total slot budget is matched. The additional nine questions recoverable by oracle allocation on the baseline decomposition in each model setting identify uneven depth requirements across arms. This result motivates evaluation of observable allocation policies. Their value must be measured after accounting for the decisions and intermediate answers needed to obtain the lists.

4.7 A Cross-Benchmark Fixed-Trace Study

4.7.1 Retrieval Regime and Evaluation Population

A complementary BM25 study applies the passage-slot accounting to MuSiQue, 2Wiki-MultiHopQA [7] and HotpotQA [6]. It uses separate closed corpora containing 11 656, 10 426 and 9 811 passages, respectively. Supporting passages are identified from each benchmark’s annotations. The metric is complete annotated-evidence availability in the selected union; it requires the evidence set but no estimated ownership mask. The ownership measurement remains relevant when the question is which arm has the evidence needed for its own decision.

Each source panel contains 1 000 questions. Saved generated-plan traces are available for 824 MuSiQue, 977 2Wiki and 996 HotpotQA questions. The full-question control is restricted to the same retained IDs within each dataset. Thus the comparisons are paired on available traces, with 176, 23 and 4 source questions outside their respective

denominators. These coverage values are conditional on obtaining a retained trace; they do not represent success rates for the entire original intake. The source panels also have historical project exposure, so these results are development evidence separate from the frozen comparison in Chapter 5.

The sequential traces use generated subquestions. A GPT-4o-mini reader answers each arm from five BM25 passages, and those predicted answers resolve later references. Up to 200 ranks per arm are saved for replay. This is a different retriever and execution implementation from the HippoRAG experiments above. Every allocation below acts on those saved lists. Its slot budget excludes the calls and contexts used to generate the intermediate answers.

Selected slots, unique pooled passages and reader tokens remain distinct. Two allocations with the same $\sum_a k_a$ can have different overlap between arms and hence different deduplicated context sizes. A subsequent reader over the pooled union is an additional policy; its answer quality must be measured separately from the coverage frontiers reported here.

4.7.2 Matched Passage-Slot Frontiers

Table 4.7 compares full-question retrieval, balanced allocation from Equation 4.13, and the fixed-list oracle at $B = 8$. The balanced allocation spends the integer remainder in execution order. The saved plans contain at most seven arms on MuSiQue and HotpotQA and six on 2Wiki, so this budget can allocate at least one slot to every arm.

Table 4.7: Complete annotated-evidence coverage at a ceiling of eight selected passage slots. BM25 results use fixed predicted-binding traces; the oracle uses gold evidence ranks. Counts are retained questions from each original 1,000-question panel.

Dataset	n	Full question	Balanced	Oracle
MuSiQue	824	0.136	0.413	0.536
2Wiki	977	0.315	0.688	0.881
HotpotQA	996	0.637	0.762	0.836

The oracle recovers 12.4, 19.3 and 7.4 additional percentage points over balanced allocation on MuSiQue, 2Wiki and HotpotQA, respectively. These are opportunities within the existing rankings. They preserve the recorded queries and reveal how unequal evidence ranks constrain a balanced split. The gain from generating better queries is outside this oracle’s policy class.

The subset dynamic program in Equation 4.15 independently reproduces the saved oracle coverage on all three retained panels at the checked budgets. For a fixed panel, its frontier is

$$C^*(B) = \frac{1}{n} \sum_{q=1}^n \mathbf{1}\{B^*(q) \leq B\}. \quad (4.16)$$

Questions whose lists jointly miss a required passage remain uncovered for every allocation. Zero-depth arms are permitted by the oracle, with the cost of constructing their saved queries kept outside the objective.

A complementary reading fixes coverage and compares the first tested budget that reaches it. On the MuSiQue sequential panel, balanced allocation first reaches coverage 0.50, 0.60 and 0.70 at 16, 32 and 128 slots. The oracle reaches the same targets at 8, 16 and 48 slots. These values are minima over the reported budget grid, rather than exact

interpolated crossings. They quantify allocation headroom and do not constitute achieved token or runtime reductions.

Generated arm count also differs from benchmark reasoning depth. Mean arm counts are 2.53, 2.79 and 2.10 on the retained MuSiQue, 2Wiki and HotpotQA panels. More arms create more allocation choices and can increase the cost of a common per-arm depth. They can also reflect redundant splitting or a prompt preference. An association between allocation headroom and arm count therefore concerns plan structure, rather than establishing that semantically harder questions obtain greater savings.

4.7.3 Reference and Generated Query Regimes

The MuSiQue artifacts support three retrieval regimes. Reference plans use the benchmark decomposition and gold intermediate answers. Sequential plans use generated subquestions and the reader’s predicted answers. Paraphrased queries replace a reference with a noun phrase derived from the referenced subquestion, without reading an intermediate answer. Table 4.8 reports their balanced coverage at two total slot budgets.

The sequential and paraphrase archives contain the same 824 question IDs and the same arm counts. Their different rankings show that the way intermediate references are turned into query text materially changes evidence availability. Reference plans provide a separate oracle-assisted comparison: they change both the source of the plan and its bindings, and their full reported panel has a different size. The three-row comparison therefore does not isolate a pure binding intervention. The controlled reference-chain comparison earlier in this chapter supplies the narrower evidence about predicted versus gold intermediate answers.

The distinction also constrains interpretation of allocation. At $B = 200$, the sequential fixed-list oracle covers 0.795 of the retained MuSiQue questions, compared with balanced coverage 0.733. Both results concern the selected prefixes of queries already produced with five passages per intermediate reader. A changed operational budget can alter those answers and later rankings, which requires re-execution to evaluate.

4.8 Discussion and Implications for Optimisation

The evaluation decomposes a retrieval plan into measurable obligations. Union depth measures pooled evidence availability. The ownership mask localises that evidence to the subquestion expected to use it. Answer metrics then measure the outcome of executing the sequence. These distinctions prevent a shallow union from being interpreted as a fully supported reasoning chain and prevent a masked-depth change caused by attribution from being attributed to improved retrieval.

Table 4.8: MuSiQue BM25 retrieval regimes. Coverage is measured on each retained panel under balanced slot allocation. The reference row uses a different plan source and a larger panel.

Query regime	n	Coverage at $B = 8$	Coverage at $B = 200$
Reference plan, gold bindings	1 000	0.595	0.962
Generated plan, predicted bindings	824	0.413	0.733
Generated plan, paraphrased references	824	0.109	0.397

The ownership study provides a working measurement model. It recovers the reference assignment on 92 of 100 complete reference-style masks while retaining high cell recall and precision. Its empirical scope is reference-style arms with known supporting passages. Generated plans can change the semantic structure through fusion, splitting, omission or incorrect dependency substitution. Chapter 5 evaluates controlled transformations of reference plans to measure sensitivity to several of these changes. Transfer to naturally generated plans requires independently labelled assignments from that regime.

The answer experiments establish the importance of execution state. Under reference decompositions, substituting gold intermediate answers changes both retrieval coverage and final accuracy. An unowned passage in the predicted mask can therefore have several explanations: a missing information need, an altered query following an earlier error, or a classifier disagreement. Feedback for optimisation should retain those distinctions. A predicted attribution diagnosis is evidence to inspect, rather than a verified causal explanation of the failure.

The resource measurements distinguish retrieval depth, total passage slots and model tokens. The masked objective rewards shallow completion of assigned evidence; the fixed-budget replay measures allocation efficiency on existing lists. A prompt that raises the former can still create more arms or consume more tokens. Chapter 5 consequently evaluates the optimisation objective together with complete-evidence coverage, answer quality and execution cost.

The complementary BM25 study extends the fixed-slot analysis to three benchmarks and larger retained panels. Its oracle gaps motivate observable allocation policies, while its query-regime comparison establishes the need to document how dependent subquestions were resolved. Those results retain their closed-corpus, fixed-trace and available-case scope alongside the separate attribution and answer measurements.

The current system applies decomposition to every question in these experiments. The difficulty probe of Chapter 3 supplies a separate measurement and is not used to select or route these executions. Connecting a pre-answer difficulty estimate to a decomposition policy requires its own cost and answer evaluation.

Conclusion

This chapter addressed the evaluation component of the retrieval-efficiency question posed in Section 2.6.2. It defined complete evidence availability and ownership-masked retrieval depth, trained a textual ownership predictor and evaluated its assignments on reference-style decompositions. The predictor recovered 269 of 274 owner cells and 92 complete question masks out of 100.

The retrieval study showed that evidence availability, assignment and answer execution vary separately. Generated decompositions increased complete pooled retrieval coverage, while the answer gains depended on the model and execution state. A fixed-budget analysis also identified room to allocate passage slots more selectively across the recorded arms. These findings support an evidence-aware optimisation objective while specifying the additional answer and cost measurements required to assess its practical effect. Chapter 5 uses this evaluation to study decomposition prompt search and learned passage allocation.

Chapter 5

Optimising Question Decomposition and Evidence Allocation

Introduction

Chapter 4 introduced an evidence-aware evaluation of a generated decomposition. The ownership mask assigns supporting passages to subquestions, and the resulting depth score measures how early those passages can be retrieved by their assigned queries. This chapter uses that score to optimise the instruction that generates the decomposition. It addresses the question posed in Section 2.6.2: can the generation strategy preserve evidence coverage while reducing retrieval depth?

The completed experiment applies an OPRO-style search to one system instruction. It evaluates twenty-four proposed instructions and the initial instruction on the same hundred MuSiQue questions. The selected instruction raises the masked depth score from 66.30 to 67.20, with unchanged exact match of 44%. These measurements describe prompt selection on the development panel. They also motivate a distinction between improving a retrieval objective and improving the answers produced by the complete system.

The chapter presents two complementary interventions. Prompt search changes the questions issued by the system. Learned allocation changes the passage prefix selected from each arm of a recorded execution. The completed allocation study uses predicted-binding BM25 traces on MuSiQue, 2Wiki and HotpotQA. At a budget of eight passage slots, its largest observed coverage increase is 6.3 percentage points on 2Wiki. These are retrieval-coverage measurements on retained traces, distinct from the pilot’s final-answer measurements. A paired 700-question MuSiQue experiment also evaluates the answers produced from balanced and segment-policy pooled contexts. The later unit-increment policy is evaluated separately on retrieval coverage.

The chapter then specifies an evidence-feedback extension that separates observed retrieval and dependency information from estimated ownership. A comparison with answer-only and information-matched raw feedback is defined on frozen optimisation, selection and confirmation splits. The extension has an implemented feedback and search interface; its comparative answer-quality evaluation remains prospective. The empirical conclusions concern the completed OPRO pilot, fixed-trace allocation study and controlled ownership stress tests.

5.1 The Optimisation Problem

5.1.1 An Instruction Defines an Executable Policy

Let p denote the decomposer’s system instruction and let q be an input question. With fixed demonstrations and generation settings, the decomposer produces an ordered plan

$$\pi_p(q) = (u_1, \dots, u_{A_p(q)}), \quad (5.1)$$

where $A_p(q)$ is the number of subquestions. A reference $\#i$ in u_a refers to the answer of an earlier arm, so valid references satisfy $i < a$. The executor substitutes its predicted intermediate answers, retrieves passages and reads them for each arm in sequence. The answer of the last arm is the final answer $\hat{y}_p(q)$.

This interface constrains useful prompt changes. Each arm must be phrased as a retrieval question, references must be executable in order, and the final arm must target the answer requested by q . A plan that retrieves relevant facts but ends with an intermediate entity can have adequate evidence coverage and an incorrect final answer. The optimisation therefore acts on a program of retrieval and reading steps, rather than on a list of paraphrases alone.

The historical experiment changes p while keeping the demonstrations, retriever, reader and ownership estimator fixed. It uses one generated plan per question. The six instructions proposed in a search round are six policy candidates, not six plans combined at inference time.

5.1.2 The Retained Retrieval Objective

For each executed plan, let $D_{M,p}(q)$ be the ownership-masked depth defined in Chapter 4. It is the greatest passage rank among the assigned arm–evidence pairs, with infinite depth when required evidence has no predicted owner or falls outside the retained ranked lists. With cutoff $K = 200$, the search objective is

$$J_M(p; \mathcal{D}) = \frac{100}{|\mathcal{D}|} \sum_{q \in \mathcal{D}} v_K(D_{M,p}(q)), \quad v_K(d) = \begin{cases} 1 - \frac{\log d}{\log K}, & 1 \leq d \leq K, \\ 0, & d = +\infty. \end{cases} \quad (5.2)$$

All questions remain in the denominator. The objective rewards early complete retrieval under the predicted assignment and gives zero credit to a censored question. Its companion coverage statistic is

$$C_M(K; p) = \frac{1}{|\mathcal{D}|} \sum_{q \in \mathcal{D}} \mathbf{1}\{D_{M,p}(q) \leq K\}. \quad (5.3)$$

Depth K receives zero logarithmic credit but counts as finite coverage. Reporting both quantities distinguishes complete evidence near the search limit from incomplete evidence.

The score depends on the ownership estimator. An assignment error can change which ranks enter the maximum, as discussed in Chapter 4. The estimator is therefore frozen during search. Evidence coverage computed directly from the gold passage set remains a separate diagnostic.

5.1.3 Answer Quality and Resource Use

The end-to-end outcomes are exact match and token F1 against the benchmark reference answers. For a fixed evaluation panel, define

$$J_{\text{F1}}(p) = \frac{1}{|\mathcal{D}|} \sum_{q \in \mathcal{D}} \text{F1}(\hat{y}_p(q), y_q). \quad (5.4)$$

The retrieval objective in Equation 5.2 and the answer objective in Equation 5.4 measure different properties. Retrieved evidence must still be read, intermediate answers must bind correctly, and the last arm must produce the requested answer.

The historical reader consumes five passages per arm. Its nominal passage-slot allowance is consequently $5A_p(q)$, with repeated passages charged to each arm that reads them. Unique passages, reader tokens and model calls are additional cost measurements. The masked-depth objective contains no direct penalty for arm count, so a higher objective value alone establishes neither lower total reading cost nor higher answer accuracy.

5.2 The Completed OPRO Search

5.2.1 Search Configuration

OPRO uses a language model to propose solutions from a history of candidates and their measured scores [4]. Here, a candidate is the text of the decomposer’s system instruction. GPT-4o proposes instructions, and the Llama-3.3-70B pipeline executes them on the development questions. Table 5.1 records the retained configuration.

The executor uses temperature-zero generation with cached model calls. Exact duplicate instruction hashes are filtered. The retained leaderboard contains twenty-five successfully evaluated candidates and zero rejected candidates. This amounts to 2 500 candidate–question evaluations. That count describes logical evaluations; cached calls can reduce the number of billable model requests.

The baseline is the seed instruction and its executions recorded within this search. The separately recorded prompt variants in Chapter 4 provide context for the pipeline, rather than substituting for this paired baseline.

Table 5.1: Configuration of the completed prompt-search pilot.

Component	Setting
Evaluation panel	The same 100 MuSiQue questions for every candidate
Initialisations	One initial instruction
Search rounds	Four, with six new instructions per round
Candidate total	25, including the initial instruction
Proposal model	GPT-4o, temperature 1.0
Execution model	Llama-3.3-70B, one plan per question
Reader context	Five passages per executed arm
Retrieval scoring	Up to 200 retained ranks per arm
Search reward	Ownership-masked logarithmic depth score
History in feedback	At most 20 prior instructions and their rounded scores
Example feedback	Three cases from the current leading instruction

5.2.2 Proposal and Selection Loop

The search begins by executing the initial instruction on all hundred questions. At each round, the optimizer receives the retained instruction history, the corresponding aggregate depth scores and three examples from the current leader. It proposes six replacement instructions. Each is executed through the same retrieval and reading pipeline, assigned an ownership mask and scored using Equation 5.2. The updated leaderboard retains the highest-scoring instruction across all completed rounds.

The full-precision score determines the leaderboard order, while the meta-prompt displays integer-rounded scores. Twelve of the twenty-five candidates are displayed in the same score-64 bucket. This reduces the resolution of the numerical information available to the proposer, even though their stored scores remain distinct.

The initial and selected instructions share the same planning policy: use complete natural-language questions, preserve independent branches, require backward references and split compound lookups into simpler arms. The selected instruction adds an explicit check that the necessary facts of the original question are covered. These are observed differences in instruction text. The experiment changes several clauses together, so the measured score change cannot be assigned to that coverage clause alone.

5.2.3 The Feedback Supplied to the Optimizer

The historical feedback prioritises questions with predicted unowned evidence, then questions with the greatest masked depth. It selects the unowned cases in stored question order. This gives the optimizer examples of poorly scored executions, but repeatedly drawing from the current leader and a fixed ordering can concentrate attention on a small part of the panel.

The example description attributes failures to facts that the plan failed to ask for. That interpretation is stronger than the underlying measurements. A passage may be assigned to an appropriate query yet appear at a large retrieval rank. Conversely, an unowned passage is an estimator output whose correctness depends on the ownership model. Both cases can lower the depth score, while suggesting different changes to the instruction. This motivates the typed feedback in Section 5.5.

5.3 Results of the Pilot

5.3.1 Progress Across Search Rounds

Table 5.2 separates the best proposal in each round from the best instruction retained so far. The overall winner appears in the second round. The third and fourth rounds produce lower-scoring candidates, while the stored best score remains 67.20.

The increase from the initial instruction is 0.9025 score points. A plateau over the final two rounds describes this trajectory and its proposal budget. Broader search, different initial instructions and alternative feedback are separate experimental choices. The four-round trajectory does not determine their attainable performance.

5.3.2 The Selected Instruction

Table 5.3 compares the selected instruction with the seed on the quantities available for both. Finite masked coverage remains at 80%. Thus the higher aggregate score is not

Table 5.2: Masked-depth score during the completed search. The retained best includes all earlier rounds. Values are on the shared development panel.

Stage	New candidates	Best in stage	Retained best
Initial instruction	1	66.3017	66.3017
Round 1	6	66.5975	66.5975
Round 2	6	67.2042	67.2042
Round 3	6	65.2557	67.2042
Round 4	6	66.0482	67.2042

accompanied by a higher aggregate count of questions with finite masked depth. Question-level successes can still change identity.

Exact match is unchanged at 44%: two questions change from incorrect to correct and two change in the opposite direction. Mean F1 increases by 0.1165 points on the 0–100 scale. The selected instruction generates 0.13 additional arms per question, corresponding to a 4.3% increase in the mean passage-slot allowance. Its measured benefit is therefore specific to the masked-depth score, with broadly similar final-answer quality and a slightly larger nominal reading allowance.

A paired bootstrap over the hundred questions gives a descriptive 95% interval of $[-1.80, 3.90]$ points for the masked-depth-score difference and $[-3.60, 4.00]$ points for the F1 difference. The exact-match difference has an interval of $[-4, 4]$ percentage points. These intervals condition on the selected pair of instructions. The same panel supplied feedback and selected the winner, so the intervals describe sensitivity to the sampled questions; they are not an independent assessment of selection generalisation.

5.3.3 The Optimisation Objective and the Answer Objective

Across the twenty-five evaluated instructions, the Spearman correlation between masked-depth score and mean answer F1 is 0.413. The candidates are adaptively related and share their evaluation questions, so this is a description of the retained candidate set. It is useful for inspecting whether the objective ranks these instructions like answer quality does.

The highest-F1 candidate occurs in the third round. It achieves F1 53.88 and exact match 46%, with masked-depth score 64.10. The depth-selected winner has F1 52.48 and exact match 44%, with score 67.20. Table 5.4 shows the resulting reversal. The highest-F1 candidate is a retrospective comparator, not a second independently selected test winner.

This result supports reporting retrieval quality and answer quality together. An

Table 5.3: Initial and selected instructions on the 100 development questions. F1 is displayed on a 0–100 scale. Passage slots are the five-per-arm allowance derived from the observed arm count.

Measurement	Initial	Selected	Change
Masked-depth score	66.3017	67.2042	+0.9025
Finite masked coverage (%)	80	80	0
Exact match (%)	44	44	0
Token F1	52.3682	52.4847	+0.1165
Mean generated arms	3.00	3.13	+0.13
Mean passage-slot allowance	15.00	15.65	+0.65

Table 5.4: Different objectives select different candidates from the same development search. The F1-leading candidate is identified retrospectively.

Instruction	Depth score	F1	Exact match (%)
Initial	66.30	52.37	44
Highest depth score	67.20	52.48	44
Highest answer F1	64.10	53.88	46

evidence-depth objective remains appropriate when the task is to bring assigned evidence earlier in the ranked lists. For a claim about improved question answering, final answers provide the direct selection criterion. Objective choice and feedback choice should therefore be varied separately.

5.4 Learned Allocation on Fixed Retrieval Traces

5.4.1 The Allocation Intervention

Prompt optimisation changes a plan and its resulting rankings. Allocation holds those rankings fixed and chooses how much evidence to retain from each arm. For a question with A recorded arms, let k_a be the prefix length selected from arm a . Its passage-slot budget is

$$\sum_{a=1}^A k_a \leq B. \quad (5.5)$$

Complete coverage means that every annotated supporting passage occurs in at least one selected prefix. This is the union criterion from Chapter 4. An occurrence in two arms is charged twice in Equation 5.5, although a pooled reader may later deduplicate it.

The learned policy is evaluated on BM25 rankings generated by a sequential execution. Each subquestion is answered using its first five retrieved passages; that predicted answer resolves references in later subquestions. The run retains 200 ranks per arm. Allocation replays these saved lists. Changing B therefore preserves the intermediate answers and downstream queries from the five-passages-per-hop trace-generation policy. The experiment measures context selection conditional on that policy, rather than a fresh budget-dependent execution of the reasoning chain.

The balanced comparator assigns $\lfloor B/A \rfloor$ slots to every arm and distributes the remainder to the earliest arms. The learned rule also starts with one slot per arm. Both comparisons use the same total budget when $B \geq A$. The retained traces contain at most seven arms, so the budgets 8, 16 and 32 reported below satisfy that condition. Zero-depth arms are allowed in the gold-informed oracle from Chapter 4; it is a reference with a broader feasible set than these floor-constrained policies.

5.4.2 Predicting Per-Arm Depth Requirements

Training assigns each supporting passage to the arm that ranks it earliest, breaking ties by arm order. An arm’s target depth d_a is the largest rank among its assigned passages, with target one for an empty assignment. If any required passage is absent from every retained list, all arm targets for that question are censored beyond the maximum depth. This convention couples the training labels across arms.

For each threshold in

$$\mathcal{K} = \{1, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 64, 96, 128, 200\},$$

a binary gradient-boosted classifier predicts

$$\hat{S}_a(k) = \widehat{\text{Pr}}(d_a \leq k \mid x_a). \quad (5.6)$$

The implementation fits separate threshold classifiers and then applies a running maximum over their predictions. This creates non-decreasing cumulative curves. It is an upward monotone envelope, with probability calibration remaining a separate property to assess.

Features describe arm text, position, arm count and the first ten BM25 scores. The score features include their spread, concentration, selected gaps and decay. They are computed from the saved rankings before prefix selection. Their acquisition is outside the passage-slot accounting in Equation 5.5. The reported budget therefore measures selected context, rather than total retrieval work, latency or model cost. On sequential traces, later resolved query text also depends on the earlier five-passage reader calls.

5.4.3 Segment and Unit-Increment Allocation

The allocation score is the separable surrogate

$$\max_{k_a \geq 1, \sum_a k_a \leq B} \sum_{a=1}^A \log \hat{S}_a(k_a). \quad (5.7)$$

It treats satisfaction of the assigned arm requirements as conditionally independent. Union coverage remains the evaluation criterion: evidence retrieved by another arm can satisfy the question even when the assigned arm misses it.

The segment policy forms an upper concave envelope of each log curve. It takes affordable whole envelope segments in descending gain-per-slot order, then distributes unused slots to arms with the largest remaining probability mass. This fills the budget, subject to the retained-depth cap. It is a heuristic for Equation 5.7, including when the original log curves are concave. For example, on depths $\{1, 2, 3\}$, log curves $(-4, -3.5, -3)$ and $(-3, -2, -1)$ with $B = 3$ produce allocation $(2, 1)$ and score -6.5 , while $(1, 2)$ achieves -6 . The remainder rule chooses the lower-confidence arm rather than the larger next marginal gain.

The unit-increment policy instead allocates one slot at a time. Let h_a be the piecewise-linear upper concave envelope of $\{(k, \log \hat{S}_a(k)) : k \in \mathcal{K}\}$. Starting from $k_a = 1$ for every arm, select an uncapped arm maximising

$$\Delta_a(k_a) = h_a(k_a + 1) - h_a(k_a) \quad (5.8)$$

and increment its depth. The retained cap is 200. Once every marginal gain is zero, additional slots can be assigned to any uncapped arm without changing the objective. With the non-decreasing predicted curves, this rule solves

$$\max_{k_a \in \{1, \dots, 200\}, \sum_a k_a \leq B} \sum_{a=1}^A h_a(k_a), \quad B \geq A. \quad (5.9)$$

Concavity makes each arm’s successive unit gains non-increasing. Every allocation corresponds to taking a prefix of those gains from each arm. Selecting the greatest available gain at each step therefore selects the greatest remaining gain overall while preserving the prefix constraint. An exchange of any smaller selected gain with a larger available gain cannot reduce the objective. Repeating this exchange gives the greedy allocation and establishes its optimality for Equation 5.9.

This exactness concerns the interpolated envelope. The envelope can lie above the original log curve even at a measured depth. For example, on depths $\{1, 2, 3\}$, curves $(-4, -3.9, -1)$ and $(-3, -2, -1)$ with $B = 3$ give unit allocation $(2, 1)$. Its original log score is -6.9 , whereas $(1, 2)$ achieves -6 . The interpolated and original objectives therefore remain distinct, and both differ from complete union coverage.

An exact finite-grid alternative is available through dynamic programming. Let $V_a(b)$ be the best surrogate score for the first a arms with at most b slots. Then

$$V_a(b) = \max_{k \in \mathcal{K}, k \leq b} \{V_{a-1}(b - k) + \log \hat{S}_a(k)\}, \quad (5.10)$$

with $V_0(b) = 0$ for $b \geq 0$ and infeasible states equal to $-\infty$. This costs $O(AB|\mathcal{K}|)$ and optimises the declared discrete surrogate. The coverage measurements below compare the segment and unit-increment policies; dynamic programming remains an alternative for the original grid objective. Surrogate optimisation and the gold-informed union oracle solve different allocation problems.

5.4.4 Evaluation Population and Cross-Validation

The saved sequential evaluations retain 824 MuSiQue, 977 2Wiki and 996 HotpotQA questions from the corresponding thousand-question panels. The missing trace counts are 176, 23 and 4, respectively. Results condition on the available traces, whose inclusion can depend on successful generation and execution. Chapter 4 records this population and the fixed-trace baselines.

Five-fold cross-validation splits by question. Every arm of a held-out question is predicted by classifiers trained on the other folds, including the feature standardisation. Shared atomic subproblems and supporting documents may recur across folds. Method choices were developed on these panels, so the resulting point estimates describe a development study rather than locked confirmation. They are distinct from the source-component-separated evaluation in Section 5.6.

Reference-decomposition results use benchmark subquestions and form a separate regime. Tables 5.5 and 5.6 use only generated sequential traces with predicted bindings. The generated-plan assignment labels are obtained from earliest passage ranks rather than an assumed alignment between arm indices and benchmark hop indices.

5.4.5 Coverage and Target Budgets

At $B = 8$, unit-increment allocation raises complete-evidence coverage from 0.688 to 0.751 on 2Wiki and from 0.413 to 0.460 on MuSiQue. The corresponding segment-policy coverages are 0.746 and 0.445. On HotpotQA, unit-increment coverage is 0.770 against the balanced 0.762 at eight slots, and falls below balanced coverage at sixteen and thirty-two slots. Exact optimisation of the envelope thus has budget-dependent effects on measured union coverage. A position-weighted allocation reaches 0.437 on MuSiQue at $B = 8$, compared with the learned segment policy’s 0.445; its corresponding 2Wiki coverage is

Table 5.5: Allocation on fixed BM25 traces. Segment denotes the segment heuristic and Unit the unit-increment optimiser of the concave interpolant. The budget counts selected slots, excluding trace generation and shallow-score acquisition. Differences are percentage points relative to balanced allocation.

Dataset	Budget	Balanced	Segment	Unit	Δ_{Unit} (pp)
MuSiQue	8	0.413	0.445	0.460	+4.7
	16	0.528	0.541	0.546	+1.8
	32	0.607	0.604	0.609	+0.2
2Wiki	8	0.688	0.746	0.751	+6.3
	16	0.864	0.884	0.892	+2.8
	32	0.926	0.932	0.932	+0.6
HotpotQA	8	0.762	0.763	0.770	+0.8
	16	0.859	0.858	0.856	-0.3
	32	0.907	0.904	0.904	-0.3

0.625. These comparators help distinguish the benefit of predicting depth from that of using a simple uneven split.

The target budgets in Table 5.6 are first crossings on the tested grid, rather than interpolated minimum costs. For example, MuSiQue reaches 50% coverage at budget 12 under segment allocation and 16 under balanced allocation. This is a selected-slot ratio of 1.33 at that target. Its interpretation excludes the common trace-generation cost and makes no assumption about final-answer accuracy.

5.4.6 Paired Final-Answer Evaluation

The completed reader experiment takes the first 700 retained MuSiQue traces and refits the segment-policy allocator using five-fold question-level cross-validation within those 700 cases. Its coverage values therefore refer to a different subset and fitted models from the 824-question frontier. Each budget-policy pair is evaluated on every selected question, giving 4 200 completed final-reader calls. These outcomes were generated before the unit-increment revision and measure the segment policy throughout.

Both policies deduplicate and interleave their selected prefixes. The same GPT-4o-mini reader, requested through the API’s model alias at temperature zero, answers the original question from that pooled context. Table 5.7 reports the paired outcomes.

At $B = 8$, segment allocation raises F1 from 0.4244 to 0.4455 and exact match from

Table 5.6: Smallest tested passage-slot budget reaching a coverage target on the fixed generated sequential traces, using the segment policy. The oracle uses gold passage ranks and permits zero-depth arms. Ratios compare discrete operating points.

Dataset	Target	Balanced	Segment	Oracle	Balanced/segment
2Wiki	0.70	10	8	6	1.25
	0.90	24	20	10	1.20
MuSiQue	0.50	16	12	8	1.33
	0.60	32	32	16	1.00
	0.70	128	96	48	1.33
HotpotQA	0.75	8	8	6	1.00
	0.90	32	32	20	1.00

Table 5.7: Pooled final-reader evaluation on 700 retained MuSiQue traces. U denotes balanced allocation and L the learned segment heuristic. All 4,200 question–budget–policy evaluations completed. The last column gives L minus U and its descriptive pointwise paired 95% F1 interval; the intervals at budgets 16 and 32 contain zero. The unit-increment allocator is a separate policy.

Budget	Cov. U	Cov. L	EM U	EM L	F1 U	F1 L	Δ F1 [95% CI]
8	0.413	0.449	0.344	0.354	0.424	0.445	+0.021 [0.004, 0.039]
16	0.523	0.547	0.360	0.377	0.454	0.464	+0.010 [−0.008, 0.027]
32	0.603	0.600	0.367	0.359	0.472	0.458	−0.014 [−0.031, 0.001]

0.3443 to 0.3543. The paired F1 difference is 0.0211, with a descriptive 95% interval of [0.0038, 0.0385] from 10 000 question-level bootstrap resamples. The corresponding F1 differences at $B = 16$ and $B = 32$ are 0.0098 and -0.0144 , with intervals $[-0.0076, 0.0270]$ and $[-0.0309, 0.0014]$. These are pointwise intervals over the recorded pairs, without adjustment across budgets or separation of shared source components. They describe the development-panel comparison rather than independent confirmation.

Every allocation spends its declared prefix budget. Deduplication leaves mean context sizes of 7.37 versus 7.39 passages at $B = 8$, and 30.04 versus 30.31 at $B = 32$. Thus the matched quantity is selected slots, with unique passages and token lengths free to differ. The observed answer gain at the smaller budget concerns this pooled-context intervention. Changing an intermediate reading budget requires regeneration of its dependent queries to test a sequential execution policy.

5.4.7 Exploratory Hidden-State Prediction of Assigned Depth

Chapter 3 studies representations of compositional hop count. A separate probe examines their association with the assigned depth target d_a defined above. The panel contains 2 648 reference arms from 1 000 MuSiQue questions with BM25 rankings. Supporting passages are assigned to the earliest-ranking arm; 31 arm targets belong to questions whose required evidence is unreachable within the retained lists. The binary targets are $d_a \leq k$ for $k \in \{2, 4, 8, 16, 32\}$.

The frozen models encode the reference subquestion templates as raw text, without a chat template. Right-padding-aware last-token pooling supplies the features at nine sampled layers per model. Of the 2 648 templates, 1 517 contain unresolved #N references and differ from the gold-resolved queries used for retrieval. The representations therefore describe the templates rather than the executed queries.

Table 5.8: Exploratory depth prediction on the fixed reference panel. Mean AUROC averages the five depth thresholds. Each model’s layer is selected using the same cross-validation results reported here.

Features	Selected layer	Mean AUROC
Qwen2.5-7B-Instruct	14	0.8397
Llama-3.1-8B-Instruct	12	0.8454
Mistral-7B-Instruct-v0.3	16	0.8442
Surface features and top-ten BM25 scores	n/a	0.7639

Five-fold cross-validation keeps all arms of a question together. Feature standardisation is fitted within each training fold. Logistic regression uses inverse regularisation

strength $C = 0.01$ for hidden states and $C = 1$ for the surface-and-score comparator, so Table 5.8 compares two specified readouts as well as their features. Near-constant hidden-state dimensions are filtered using the full panel before splitting.

The split retains substantial source overlap: 1 557 test-arm instances have an identical template in their training fold, and 755 test questions share a source subproblem with training. Layer selection on the reported folds adds selection optimism. All selected-layer activations are finite; the Qwen final-layer arrays contain nonfinite activations for 2 237 arms, which the probe code replaces with zero. That layer is unsuitable for interpreting a layer-wise performance decline.

These measurements motivate a component-separated evaluation of template representations with nested layer selection and matched readout tuning. Their use in an allocator requires a further end-to-end comparison of evidence coverage, answer quality and representation-extraction cost.

5.5 Evidence-Aware Feedback for Prompt Search

5.5.1 Separating Reward from Diagnostic Information

A scalar reward ranks candidate instructions. A diagnostic record identifies features of their executions that the optimizer can use when proposing a revision. Keeping these roles separate allows the same final-answer reward to be paired with different feedback representations.

For a question q , the proposed record contains the generated plan, resolved queries, intermediate answers, retrieved evidence and final-answer score. The gold passages and reference decomposition are available to the optimisation-side feedback service. During execution, the decomposer and reader use the question, fixed demonstrations, retrieved passages and their own predicted bindings. Gold intermediate answers enter feedback after the execution rather than resolving its dependencies.

The implemented feedback builder supports three views:

1. **Answer feedback.** The question, generated plan, final answer, reference answers, exact match and F1.
2. **Raw evidence feedback.** Answer feedback plus reference obligations and their passage text, executed queries, intermediate predictions and passage ranks. Available cell votes from the frozen ownership ensemble are included as measurements.
3. **Structured evidence feedback.** The same raw evidence payload plus explicit dependency edges, per-obligation retrieval status, estimated ownership status and restricted binding checks.

The raw and structured views share identical underlying evidence text and ownership measurements for a given trace. Their shared payload is hashed after common clipping, and both obey the same overall reflection input cap. Structured feedback adds a derived organisation of that information. Its comparison with raw feedback therefore tests the value of the representation, while the comparison with answer-only feedback also includes additional supervision. Actual reflection tokens remain a reported cost because equal caps do not imply equal realised input lengths.

5.5.2 Observed Events and Estimated Assignments

Table 5.9 distinguishes deterministic observations from classifier-dependent interpretations. The distinction prevents a failed answer from being assigned a single cause on the basis of a score alone.

The current binding check uses normalised exact subquestion templates and requires their referenced predecessors to align consistently. It leaves paraphrased, merged or otherwise ambiguous arms unmatched. A benchmark answer mismatch on an aligned arm is evidence about that intermediate prediction, not proof that it caused the final error. Causal attribution would require controlled re-execution that changes the suspected failure while holding the remaining policy fixed.

Ownership estimates retain the five model votes and the identity of the frozen estimator. A predicted unowned passage is described as a suspected unassigned obligation. When a passage is available in a different arm’s context, the record distinguishes global availability from the evidence available to its estimated owner. This is the distinction that the union and masked depths formalised in Chapter 4.

5.5.3 Reflective Search with a Fixed Execution Interface

GEPA supplies a reflective prompt-search procedure that uses execution feedback and candidate performance across examples [5]. The extension uses the pinned GEPA implementation to optimise the single decomposer instruction. The same adapter evaluates answer, raw and structured feedback conditions. The proposed contribution is the evidence-feedback representation within this common search procedure.

The primary comparison uses final-answer F1 as its scalar reward in every condition. The original masked-depth score, complete-evidence coverage and resource use are retained as diagnostics. Candidate selection uses the instance-level Pareto strategy, with reflection batches of eight optimisation examples and strict improvement acceptance. Examples are sampled by shuffled epochs, and their exposure counts are logged.

Only one text component is optimised. The component-exchange merge operator is disabled, because it would not provide a substantive combination of different components in this setting. The experiment is consequently a single-component reflective mutation comparison. Calling it an evolutionary search does not imply that every possible mutation, crossover or weight-training method is evaluated.

The execution contract permits one to six arms and strictly backward references. Invalid plans trigger a shared direct-question fallback, which remains in the quality and cost denominators. The reader uses five passages per arm, and the same model settings, demonstrations and passage corpus are used across fresh search conditions. This matches the maximum execution allowance while leaving actual arm count and tokens free to differ.

Fresh task execution uses GPT-4o-mini, snapshot `gpt-4o-mini-2024-07-18`. Reflection uses GPT-4o, snapshot `gpt-4o-2024-08-06`. The runner checks both returned model identifiers. The declared task seed and first search seed are one, with temperature zero for execution and one for reflection. Provider identity and generation settings enter the cache and run fingerprints. These settings remain separate from the historical Llama-based pilot.

Table 5.9: Diagnostic fields supplied by the structured feedback interface.

Field	Measurement	Scope
Dependency structure	Plan references and interface validity	A property of the generated plan
Reader evidence gap	Required passage absent from every five-passage reader context	An observation of this execution
Retained-list gap	Required passage absent from all saved rankings	Censored at the retained depth
Unassigned obligation	An obligation with zero estimated owners	An estimated assignment gap
Fold disagreement	Votes differ for an arm–evidence pair	Estimator uncertainty
Binding comparison	Intermediate answer compared with an aligned reference	Available only for a uniquely matched template and its dependencies

5.6 Independent Evaluation Design

5.6.1 Frozen Data Roles and Corpus Scope

The new comparison separates questions used for reflection, questions used for candidate selection and questions reserved for final confirmation. The source is the official answerable MuSiQue development release [1]. Excluding all composite question IDs in the historical thousand-question panel leaves 1 417 cases. Removing three additional exact normalised question-text matches leaves 1 414.

Of those questions, 363 have every gold supporting passage in the fixed 11 656-passage corpus. The initial matched-backbone study uses this eligible subset. Eligibility is determined before new model outputs, by exact passage identity rather than observed retrieval success. It defines a corpus-covered target population. Broader MuSiQue performance requires a separate evaluation with the additional corpus material included.

Questions sharing any underlying single-hop source ID are grouped before the new splits are assigned. The final manifests contain 96 optimisation, 48 selection and 200 confirmation questions, with zero shared source IDs across these three splits. Table 5.10 gives their composition. Split identities and file hashes are frozen before candidate generation.

All selected composite question IDs and texts differ from the historical panel and the ownership classifier’s training panel. Historical subproblem reuse remains: 343 of the 344 selected questions share at least one single-hop source with the historical thousand questions. These are new compositions under the documented exposure exclusions, rather than wholly new atomic facts. There is no source-ID overlap with the classifier’s training

Table 5.10: Frozen question-new splits for the prospective comparison. Components connect questions sharing a source single-hop ID.

Role	Questions	Two-hop	Three-hop	Four-hop	Components
Optimisation	96	23	40	33	29
Selection	48	16	19	13	22
Confirmation	200	73	77	50	65

questions. The confirmation set therefore tests transfer to new compositions within the corpus-covered regime.

The reflection service accepts only hash-bound optimisation samples. The selection set supplies scores for ranking candidates and remains adaptive development data. Confirmation question contents and labels stay outside the search interface until the final instructions and analysis choices have been fixed.

5.6.2 Comparators and Search Allowances

The fresh study includes the unchanged initial instruction, nonadaptive instruction sampling, OPRO-style answer-feedback search and three GEPA feedback conditions. Non-adaptive proposals are generated from the task and initial instruction before their evaluation results are available. The OPRO control uses score histories and answer errors. The three GEPA conditions share the optimizer, reward and execution interface while varying the feedback view.

Each fresh search is assigned a ceiling of 512 example-level rollout evaluations and 24 candidate proposals. A rollout is an executed candidate-question pair. Parent comparisons, proposed-candidate comparisons and validation executions count toward that ceiling. The different methods allocate their allowance differently, so search progress is measured against cumulative evaluations and actual cost.

The nonadaptive schedule screens 24 instructions on a common set of 12 optimisation questions, then evaluates its four leaders on the 48 selection questions. This uses

$$24 \times 12 + 4 \times 48 = 480 \quad (5.11)$$

candidate-question evaluations. The OPRO control uses four rounds of six proposals, with a declared twelve-question screen per round and selection evaluation for each round's nominee. The reflective search uses its allowance for parent and child batches and complete selection evaluations. The number of completed mutations is reported explicitly rather than inferred from the evaluation ceiling.

Within a search condition, the winner is selected by mean selection F1. Ties are resolved by lower mean execution tokens and then instruction hash. The fixed initial instruction and the historical OPRO winner are retained as identified reference policies. The fresh pipeline may use a different API model configuration from the historical Llama pilot; comparisons are made within a frozen fresh backbone, and historical scores remain in their original regime.

The principal contrast is structured versus answer-only feedback. Structured versus raw feedback is required to isolate the effect of diagnostic organisation from additional supervision. A second predeclared search seed provides a replication of search variability when resources permit. Both seeds' trajectories must then be retained.

5.6.3 Outcome Analysis and Cost Accounting

After selection, each final policy is evaluated on the same locked confirmation questions. The primary outcome is the question-weighted mean paired F1 difference. Exact match, complete evidence in the reader contexts, union depth, masked depth, arm count and model tokens are secondary measurements. All questions, including fallback executions, contribute to the aggregate results.

Paired bootstrap resampling preserves each question’s outcomes across policies. A complementary component-level bootstrap keeps questions sharing a source component together. The confirmation panel contains 65 such components, with at most 20 questions in one component. Question and component resampling describe sensitivity under different sampling units; neither supplies a distribution-free generalisation guarantee.

Search cost and inference cost are recorded separately. Search cost includes instruction proposals, reflection and candidate evaluation. Inference cost includes decomposition, retrieval-associated model calls and every reader step for the selected policy. Cached deterministic evaluations count toward the logical search allowance while their actual charges are recorded separately. A cache hit is reuse of a result, not an independent execution.

The passage-allocation replay in Chapter 4 provides a complementary cost diagnostic. On recorded lists it compares balanced total-slot allocation with the smallest oracle prefix allocation covering the gold passages. The oracle has access to gold identities and frozen rankings. It measures available allocation headroom; it is not a learned execution policy. In the sequential pipeline, changing an early reader allowance may change its answer and every downstream query. Such a policy requires fresh execution before claiming an answer-quality or runtime gain.

5.6.4 Separating Prompt and Allocation Effects

The completed studies intervene at different points: the OPRO pilot changes the decomposition instruction, while Section 5.4 changes prefixes selected from fixed rankings. A prospective factorial comparison can test their combination. Freeze an initial prompt p_0 , a search-selected prompt p_1 , balanced allocation g_0 and learned allocation g_1 before final evaluation. Evaluate all four pairs (p_i, g_j) on the same questions and reader budget. If Y_{ij} denotes mean final-answer F1, the interaction is

$$I = (Y_{11} - Y_{10}) - (Y_{01} - Y_{00}). \quad (5.12)$$

It asks whether the allocation benefit changes under the new prompt. Each prompt generates its own traces under the same five-passages-per-hop policy; both allocators then act on those traces. This comparison concerns pooled context, with trace-generation cost reported separately. Its outcomes remain prospective.

A sequential extension must choose each arm’s budget using information available at that step and reserve resources for later arms. It then recomputes intermediate answers and downstream retrieval. The offline allocator uses resolved later queries and their score profiles, which would give an online policy future information. Step-local allocation and re-execution are therefore part of that extension.

5.6.5 Controlled Checks of the Diagnostic Representation

A prepared stress suite examines how ownership diagnostics should respond to known plan transformations. Twelve optimisation questions are selected deterministically before classifier evaluation, with four at each of two, three and four hops and twelve distinct source components. They yield 52 cases: 12 reference plans, 12 relevant-arm duplications, 12 irrelevant-arm insertions, 12 sink-arm omissions and 4 independent-arm fusions.

Expected ownership masks are derived from transformation provenance. Duplication adds an owner for the same supporting passage. An irrelevant arithmetic query adds an

arm with no obligation from the reference task. Omitting an unreferenced sink creates a missing obligation only when its supporting passage is unique among the remaining arms. Shared evidence is tracked separately so that deleting a hop is not automatically equated with deleting its passage coverage.

The four fusions combine independent reference arms whose distinct answers are subsequently used downstream. The scalar reference grammar provides no selector for multiple outputs from one fused arm. These four cases therefore use explicitly gold-resolved downstream inputs and test ownership on synthetic plans. They are excluded from claims about executable fused policies. The remaining transformations preserve valid backward references.

The frozen five-model ensemble was evaluated on all 52 cases using the reconstructed feature pipeline. Table 5.11 compares the predicted mask with the complete provenance-derived mask for each case. These transformation-suite measurements are separate from the historical classifier evaluation in Chapter 4.

Reference plans and relevant-arm duplications preserve the expected masks in all twelve cases. After sink omission, the ensemble identifies six of the twelve missing obligations. Irrelevant insertion and omission produce additional positive assignments, making missingness sensitive to the estimator. This supports describing an unowned obligation as suspected rather than treating the mask as an omission oracle. The suite measures response to controlled, reference-derived transformations; the variants are dependent and the fused cases retain their gold-resolved scope.

5.7 Discussion

5.7.1 What the Completed Experiment Establishes

The pilot uses the attribution method of Chapter 4 within an executable prompt search. It demonstrates that the measured instruction objective can change under automated proposals while the retrieval and reading backbone remains fixed. The selected instruction improves the masked-depth score by 0.90 points on the development panel, preserves aggregate finite masked coverage and produces similar final-answer quality. Its higher mean arm count makes the cost distinction explicit.

The candidate-level comparison also identifies a practical choice. Selecting for evidence depth and selecting for final-answer F1 can choose different instructions from the same search. An optimiser intended to improve QA should therefore be evaluated on the answer outcome, with the evidence measurements explaining where execution changes occur. An optimiser intended to improve retrieval depth should state that objective and measure

Table 5.11: Ownership predictions on the optimisation-only stress suite. Exact agreement requires every cell of a case’s ownership mask to match the transformation labels. Variants share twelve base questions.

Transformation	Cases	Exact masks
Reference plan	12	12
Duplicate relevant arm	12	12
Insert irrelevant arm	12	7
Omit sink arm	12	6
Fuse independent arms, gold-resolved	4	2

total resource use alongside it.

The allocation study adds a measured intervention on the evidence selected from recorded rankings. Its coverage gains are largest on 2Wiki at the smaller tested budgets, with more limited differences on MuSiQue and HotpotQA. The unit-increment policy optimises the concave interpolant exactly. Prediction, assigned-depth targets, that interpolant and the arm floor still contribute to the measured policy. The gap to the gold-informed union oracle combines these factors. The paired segment-policy MuSiQue reader experiment shows a 2.1-point F1 increase at eight slots and a 1.4-point decrease at thirty-two slots. Allocation therefore needs assessment at the intended budget and on final answers, alongside the evidence-coverage objective.

5.7.2 The Scope of the Evidence-Feedback Extension

The extension turns the rank matrix and ownership estimates into structured records for prompt revision. Its design separates an observed reader-context gap, an estimated assignment gap and an aligned intermediate-answer mismatch. It also preserves an information-matched raw view, enabling a direct test of whether this organisation helps the optimizer.

Chapter 4 describes the ownership model’s benchmark evaluation. The stress suite adds controlled duplication, insertion, omission and fusion measurements. Automatically optimised instructions can generate other arm styles and relations, which require a generated-plan attribution audit. Keeping final-answer F1 as the primary reward reduces reliance on estimated ownership for winner selection, while preserving ownership as diagnostic feedback.

The fresh comparison is designed to establish performance of frozen instructions on new compositions. The fixed corpus eligibility filter, historical subproblem reuse and finite confirmation panel delimit its generalisability. Transfer to an expanded corpus, another dataset or another reader is a separate experiment. The chapter supplies completed prompt and allocation studies and a controlled protocol for measuring whether evidence feedback and allocation improve final-answer quality together.

Conclusion

This chapter optimised a decomposition instruction using the ownership-masked retrieval score introduced in Chapter 4. Four rounds produced twenty-four new instructions, evaluated alongside the seed on one hundred questions. The selected instruction increased the score from 66.30 to 67.20, while exact match remained 44% and F1 changed from 52.37 to 52.48. The distinction between retrieval depth, answer quality and total execution cost is therefore retained in the assessment of prompt changes.

A complementary learned allocator uses arm features and shallow score profiles to distribute a fixed passage-slot budget. On generated sequential BM25 traces, unit-increment allocation increases eight-slot coverage by 6.3 percentage points on 2Wiki, 4.7 on MuSiQue and 0.8 on HotpotQA. It optimises the concave-interpolant objective exactly; its measured coverage effects vary across budgets. Their cost accounting separates selected passages from the computation that generated the traces. On the earlier segment policy’s 700-question pooled-reader comparison, F1 rises from 0.424 to 0.445 at eight slots, with budget-dependent effects across the three tested settings.

The proposed continuation uses final-answer F1 for selection and evidence responsibility for diagnostic feedback. Its implemented interfaces separate raw observations from

estimated assignments, and its frozen evaluation design compares answer-only, raw and structured feedback under common search and execution allowances. Together, Chapters 4 and 5 provide an evidence-aware measurement, its use in completed prompt and allocation studies, and a controlled design for testing their contribution to answer quality on new questions.

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